CS 391
Web Application Development

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Week - 7

Outline

1. Components
2. Props
3. Component State

React Core Concepts

- There are three core concepts of React:
  - Components
  - Props
  - State

Components

- User interfaces can be broken down into smaller building blocks called components. Like **LEGO bricks**
- Components allow you to build self-contained, reusable snippets of code.
- A component is a function that returns UI elements. Inside the return statement of the function, you can write JSX

Simple Component Examples

```
const Hello = () => {
    return <div> <b>My First Hello Component</b> </div>
 }
export default Hello;
```

```
export default function About(){
    return <div>About Page</div>
}
```

```
const Hello = () => {
    return (<div>
              <b>My First Hello Component</b>
          </div>)
}
export default Hello;
```

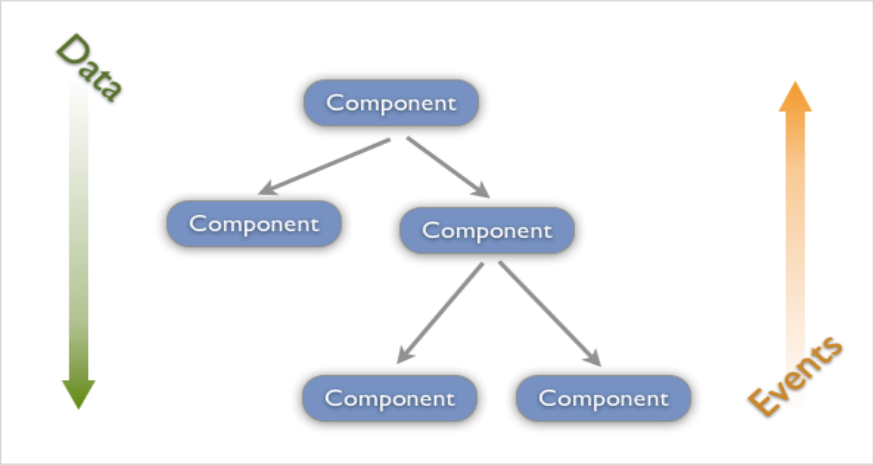
Without parentheses, any code on the lines after return will be ignored!

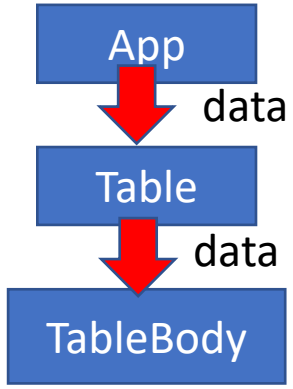
Component Properties

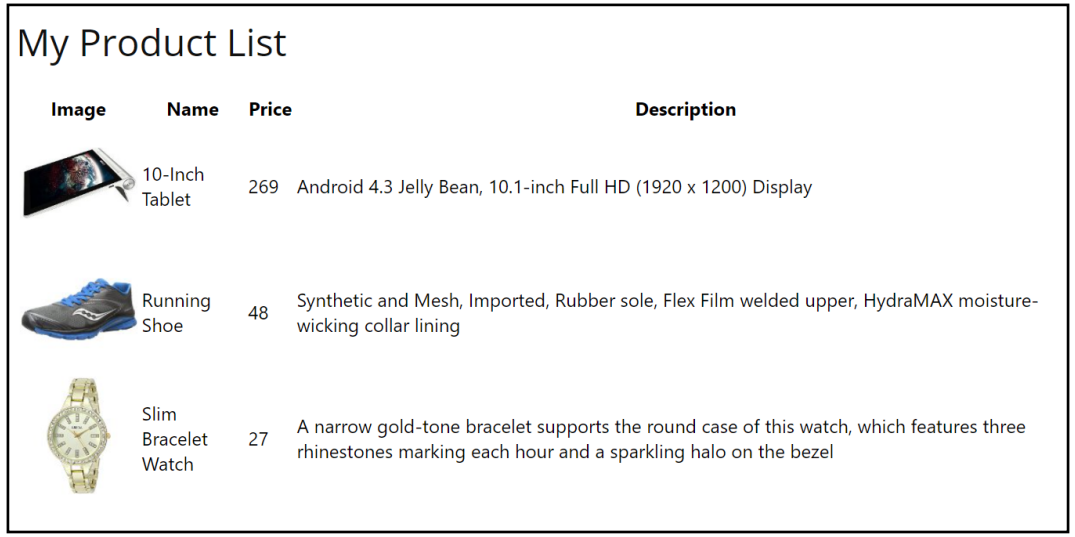
- Props are
  - arguments passed into React components.
  - passed to components via HTML attributes
- Accessible with **props**
- Props are read-only for child elements.

Component Oriented Development

- Data should always flow down







My Product List

Image	Name	Price	Description
	10-Inch Tablet	269	Android 4.3 Jelly Bean, 10.1-inch Full HD (1920 x 1200) Display
	Running Shoe	48	Synthetic and Mesh, Imported, Rubber sole, Flex Film welded upper, HydraMAX moisture-wicking collar lining
	Slim Bracelet Watch	27	A narrow gold-tone bracelet supports the round case of this watch, which features three rhinestones marking each hour and a sparkling halo on the bezel

Example…

data.js

```
export const data = [{...},{...},...]
```

- App.js

```
import {data} from './data.js'
import FilteredTable from './components/FilteredTable';
function App() {
  return (
    <div className="App">
        <FilteredTable products={data}></FilteredTable>
    </div>
  );
}
export default App;
```

…Example

FilteredTable.js

```
const FilteredTable = (props) => {
    const { products} = props;
    return (<table>
                <tr><th>Name</th><th>Price</th><th>Description</th></tr>
                 {products.map((item, index) => {
                     return (<tr key={index}>
                                 <td>{item.name}</td>
                                 <td>{item.price}</td>
                                 <td>{item.description}</td>
                             </tr>
                            );})}
            </table>
    );
}
export default FilteredTable;
```

This syntax is called “destructuring”

…Example

- Usually you don't need the whole props object itself, so you destructure it into individual props.

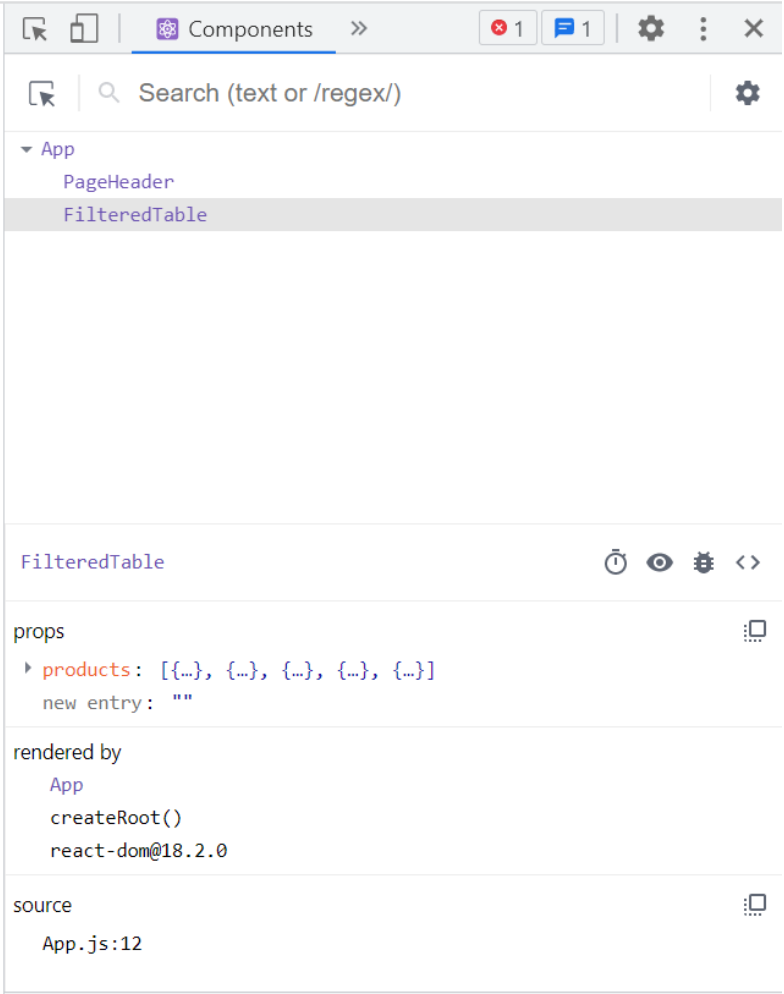
reading properties from a function parameter

```
const FilteredTable = ({products}) => {
        return (<table>
                <tr><th>Name</th><th>Price</th><th>Description</th></tr>
                 {products.map((item, index) => {...})}
            </table>
    );
}
export default FilteredTable;
```

Sample Output

my header

Name	Price	Description
Laptop Stand	$49.99	Laptop Stand for Desk, Adjustable Computer Stand with 360° Rotating Base, Ergonomic Laptop Riser for Collaborative Work, Foldable & Portable Laptop Stand, fits for All 10-17" Laptops(Grey
Curved Computer Monitor	$149.99	Curved Computer Monitor, AMD FreeSync for Advanced Gaming, 4ms Response Time, Wide Viewing Angle, Ultra Slim Design, LC24F396FHNXZA, Black
Wireless Internet Router	$154.99	TP-Link AX5400 WiFi 6 Router (Archer AX73)- Dual Band Gigabit Wireless Internet Router, High-Speed ax Router for Streaming, Long Range Coverage
Bluetooth Earbuds	$139.99	Jabra Elite 7 Pro in Ear Bluetooth Earbuds - Adjustable Active Noise Cancellation True Wireless Buds in a Compact Design with Jabra MultiSensor Voice Technology for Clear Calls - Titanium Black
Laptop Backpack	$22.94	Monsdle Travel Laptop Backpack Anti Theft Water Resistant Backpacks School Computer Bookbag with USB Charging Port for Men Women College Students Fits 15.6 Inch Laptop (Purple)

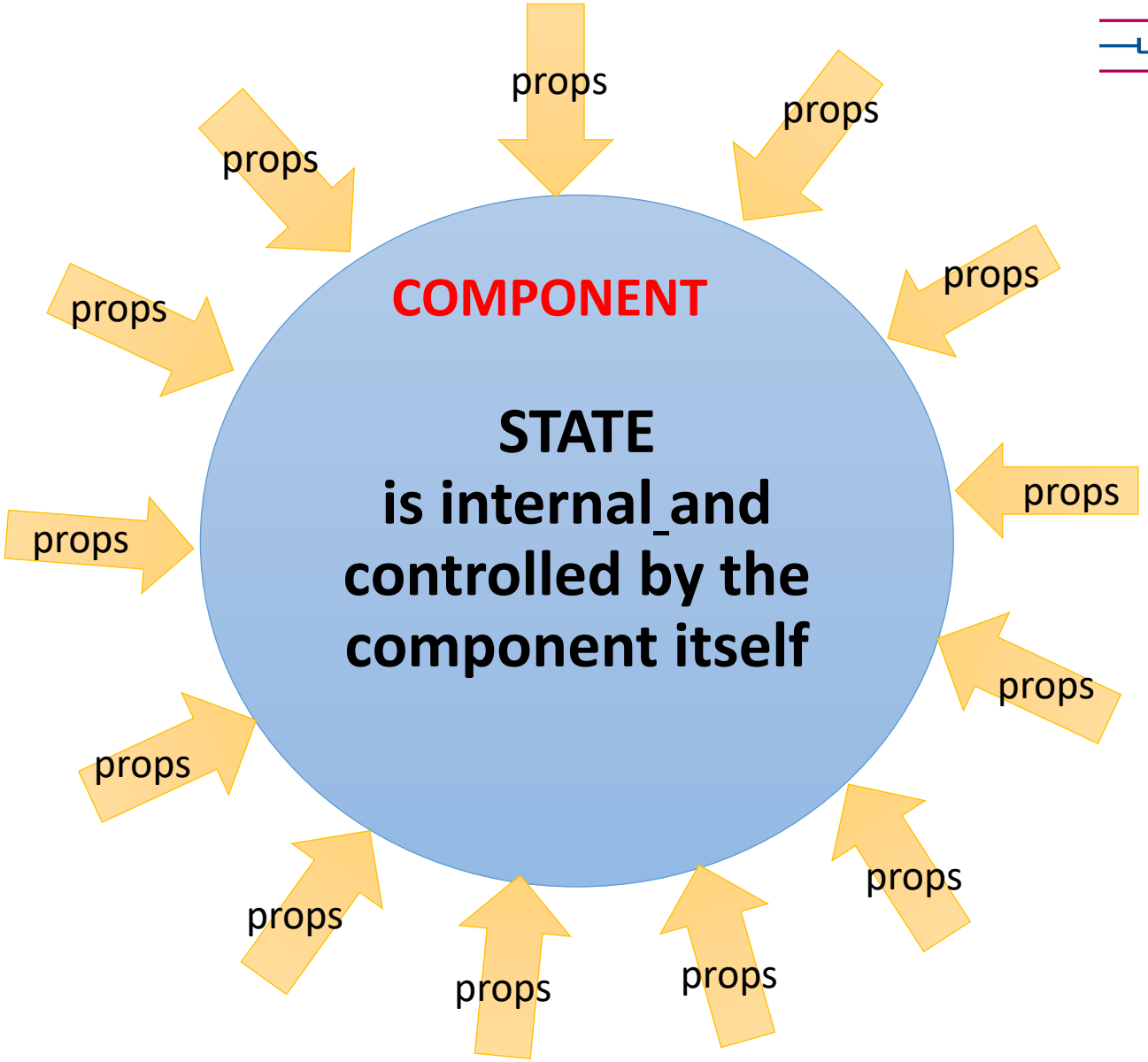


React State

- The **state** object is where you store property values that belongs to the component.
- When the state object changes, the component re-renders.

State & Props

PROPS
are external and controlled by whatever renders the component



Which data is state?

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Simply ask these questions about each piece of data:

- Is it passed in from a parent via props? if so, then it's not a state
- Does it change over time? if so, then it's a state

Updating the screen

- To do this, add state to your component
- First, import useState from React:
  - import { useState } from 'react';
- Declare a state variable inside your component:
  - const [count, setCount] = useState(0);
  - count: the current state (count)
  - setCount: the function that lets you update it

...State Example

```
import { useState } from 'react';

const Counter = () => {
    const [count, setCount] = useState(0);

    return (
        <div>
            <p>{count} times clicked</p>
            <button onClick={() => setCount(count + 1)}>
                Click to increment
            </button>
            <button onClick={() => setCount(count - 1)}>
                Click to decrement
            </button>

        </div>
    );
}
export default Counter;
```

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Click to increment | Click to decrement

Important Notes…

- Do Not Modify State Directly.

```
// Wrong
count = 7;
```

```
// Correct
setCount({ count+1});
```

Updating Arrays in State…

- In JavaScript, arrays are just another kind of object.
  - Like with objects, you should treat arrays in React state as read-only.
  - This means that you shouldn’t reassign items inside an array like arr[0] = 'bird', and you also shouldn’t use methods that mutate the array, such as push() and pop()
- Just like with objects, when you want to update an array stored in state, you need to create a new one (or make a copy of an existing one) and then set state to use the new array.
- you can create a new array from the original array in your state by calling its non-mutating methods like filter() and map()

Updating Arrays in State

- Here is a reference table of common array operations

	avoid (mutates the array)	prefer (returns a new array)
adding	`push`, `unshift`	`concat`, `[...arr]` spread syntax
removing	`pop`, `shift`, `splice`	`filter`, `slice`
replacing	`splice`, `arr[i] = ...` assignment	`map`
sorting	`reverse`, `sort`	copy the array first

For more info : https://react.dev/learn/updating-arrays-in-state

ES6 Spread Operator

- The spread operator (...) in JavaScript provides a simple and expressive way to expand elements from arrays, strings, or objects.
- It helps make code cleaner by reducing the need for manual copying or looping.

Spread Operator Examples

```
const numbers = [1, 2, 3];

const copyNumbers = [...numbers];

console.log(copyNumbers);  // [1, 2, 3]
```

```
// expand using spread operator

let a = [10, 20];
let b = [...a, 30, 40];

console.log(a); // [10, 20, 30, 40]
```

```
// expand using spread operator

let a = [10, 20];
let b = [30, 40, ...a, 50, 60];

console.log(b); //  [30, 40, 10, 20, 50, 60]
```

```
// Spread operator for array concatenation
let a = [1, 2, 3];
let b = [4, 5];

a = [...a, ...b];
console.log(a); // [1, 2, 3, 4, 5]
```

Example: Adding new element

```
export default function List() {
  const [name, setName] = useState('');
  const [artists, setArtists] = useState([]);
  return (
    <>
      <input value={name} onChange={e => setName(e.target.value)}/>
      <button onClick={() => {
        setArtists([...artists, { id: nextId++, name: name }]);
      }}>Add</button>
      <ul>
        {artists.map(artist => (<li key={artist.id}>{artist.name}</li>))}
      </ul>
    </>
  );
}
```

Spread syntax

Example: Updating an element

```
const likeHandler = (id) => {
    const array = productList.map((product) => {
      if(product.id==id){
        product.like++;
        return product;
      }else{
        return product;
      }
    });
    setProductList[array];
  }
```

Any Question?

Thank you for your attention

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